The No-U-Turn Sampler

Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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MCMC & Hamiltonian Monte Carlo

- Exact posterior inference is often intractable; MCMC is asymptotically unbiased [Nea93].
- Random-walk & Gibbs [Met+53; GG84] struggle in high-dimensional or correlated spaces.

Hamiltonian Monte Carlo

Auxiliary momentum $r \sim \mathcal{N}(0, I) +$ leapfrog: $p(\theta, r) \propto \exp\{\mathcal{L}(\theta) - \frac{1}{2} r^\top r\}$. Volume-preserving $\Rightarrow$ proposals can be **distant** yet **highly probable**.

Good L requires costly preliminary runs and expert knowledge, a key barrier to adoption.

Step size ε

Too small $\Rightarrow$ wasted computation

Too large $\Rightarrow$ high rejection rate

Leapfrog steps L

Too small $\Rightarrow$ random walk

Too large $\Rightarrow$ trajectory loops back

NUTS addresses both tuning problems simultaneously.

NUTS Methodology

No-U-Turn criterion

Stop when the trajectory doubles back: $(\theta^+ - \theta^-)\cdot r^- < 0$ or $(\theta^+ - \theta^-)\cdot r^+ < 0$.

Recursive doubling

1. Draw direction $v_j \sim \mathrm{Unif}(\{-1, +1\})$
2. Take 2^j leapfrog steps of size $v_j \varepsilon$
3. Grow balanced binary tree; halt on U-turn

Memory $\mathcal{O}(2^j) \to \mathcal{O}(j)$: partition $\mathcal{C} = \mathcal{C}^{\mathrm{old}} \cup \mathcal{C}^{\mathrm{new}}$, propose $w' \in \mathcal{C}^{\mathrm{new}}$, accept with $\min\left(1, \frac{|\mathcal{C}^{\mathrm{new}}|}{|\mathcal{C}^{\mathrm{old}}|}\right)$.

Dual averaging (warm-up only)

Drive acceptance to target δ:

$$x_{t+1} = \mu - \frac{\sqrt{t}}{\gamma(t + t_0)} \sum_{i=1}^{t} H_i,$$

$$\bar{x}_{t+1} = t^{-\kappa} x_{t+1} + (1 - t^{-\kappa}) \bar{x}_t$$

$x = \log \varepsilon$, $\mu = \log(10\varepsilon_0)$

Fixed Hyperparams: $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$

No-U-Turn criterion + dual averaging $\Rightarrow$ NUTS is **entirely tuning-free**.

Evaluating NUTS: The Paper's Benchmarks

The paper tests NUTS on four **unimodal or mildly structured** targets.

250-D	25-D	302-D	3001-D
Multivariate Normal	Logistic Regression	Hierarchical LR	Stochastic Volatility

NUTS matches or exceeds best-tuned HMC on **all four benchmarks** without any trajectory-length tuning. What about posteriors with complex geometry?

Multimodality

BNN, 191-D

$5! \times 5! =$

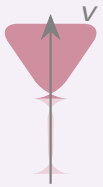
14400 modes

Dense Correlations

GP Classif., 150-D

dense 150×150 cov.

Varying Geometry



Neal's Funnel, 10-D

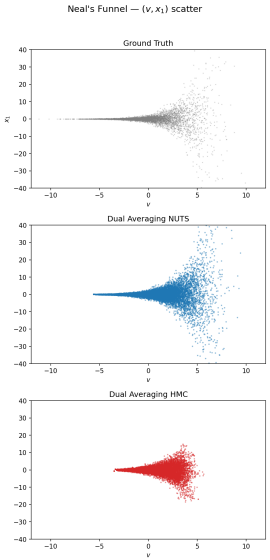
spatially varying σ

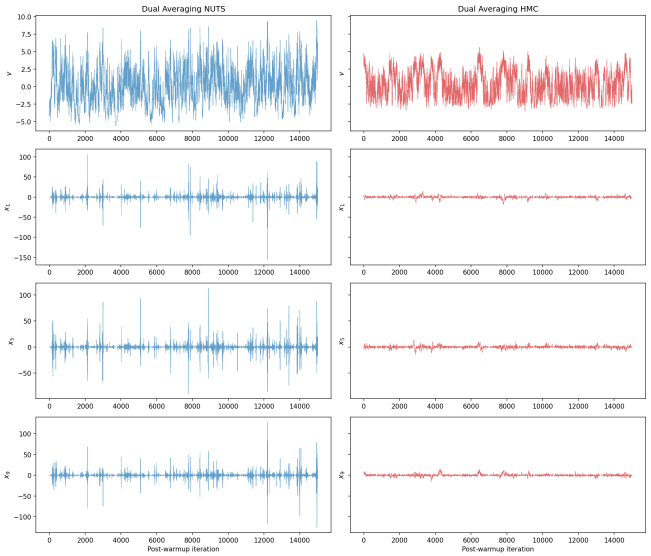
Beyond the Paper: Three Stress Tests

Results

- **BNN:** NUTS explores wider weight space but lower ESS/grad than well-tuned HMC (0.03 vs. 1.57×10^{-4}).
- **GPC:** Modest 40% ESS/grad gain, but NUTS uses $4\times$ more gradient evaluations; advantage is narrow.
- **Funnel:** NUTS recovers $\mathrm{Var}(v) \approx 9$ (truth); HMC's fixed ε fails to explore the neck.

Adaptive trajectory length is critical on complex posteriors.





Summary & Contributions

Paper assessment

+ Resolves HMC's central practical limitation; conceptually simple yet theoretically rigorous
+ No-U-Turn criterion + recursive doubling preserve detailed balance
+ Dual averaging for step-size adaptation $\Rightarrow$ essentially tuning-free
− Dual-averaging hyperparameters justified empirically; robustness across geometries is open
− Performance guarantees mainly asymptotic; limited range of posterior geometries tested

Our contributions

- **Proofs** of uniform $\mathcal{C}$ distribution and detailed balance of the memory-efficient kernel (only sketched in paper)
- **Reproduced** all 4 benchmarks; NUTS matches or exceeds best-tuned HMC without trajectory-length tuning
- **Stress tests:** adaptive trajectory length is critical on complex posteriors, the gains are most dramatic on Funnel & BNN; advantage narrows on dense GP
- **Conclusion:** confirms NUTS's suitability as a general-purpose inference engine for modern Bayesian models

Bibliography i

[Met+53] Nicholas Metropolis et al. **"Equation of state calculations by fast computing machines".** In: *The journal of chemical physics* 21.6 (1953), pp. 1087–1092.

[GG84] Stuart Geman and Donald Geman. **"Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images".** In: *IEEE Transactions on pattern analysis and machine intelligence* 6 (1984), pp. 721–741.

[Nea93] Radford M Neal. **"Probabilistic inference using Markov chain Monte Carlo methods".** In: (1993).

Appendix: The Transition Kernel & Detailed Balance

To reduce memory from $\mathcal{O}(2^j)$ to $\mathcal{O}(j)$, NUTS partitions the candidate set: $\mathcal{C} = \mathcal{C}^{\text{old}} \cup \mathcal{C}^{\text{new}}$.

The Proposal:

- From state $w \in \mathcal{C}^{\text{old}}$, propose a jump to $w' \in \mathcal{C}^{\text{new}}$ uniformly.
- Accept with probability: $\min\left(1, \frac{|\mathcal{C}^{\text{new}}|}{|\mathcal{C}^{\text{old}}|}\right)$

Proof of Detailed Balance (Intuition)

We must satisfy: $p(w|\mathcal{C})T(w'|w,\mathcal{C}) = p(w'|\mathcal{C})T(w|w',\mathcal{C})$

Substituting the prior $p(w|\mathcal{C}) = \frac{1}{|\mathcal{C}|}$ and the acceptance rates:

$$\frac{1}{|\mathcal{C}||\mathcal{C}^{\text{new}}|} \min\left(1, \frac{|\mathcal{C}^{\text{new}}|}{|\mathcal{C}^{\text{old}}|}\right) = \frac{1}{|\mathcal{C}||\mathcal{C}^{\text{old}}|} \min\left(1, \frac{|\mathcal{C}^{\text{old}}|}{|\mathcal{C}^{\text{new}}|}\right)$$

Proving the kernel preserves the target distribution.

Appendix: Uniform Distribution over the Candidate Set

Claim: $p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) = \frac{1}{|\mathcal{C}|}\, \mathbb{I}[w \in \mathcal{C}]$

Let $w = (\theta, r)$. The augmented joint density is
$p(\theta, r, u) \propto \mathbb{I}\big[u \leq \exp\big(\mathcal{L}(\theta) - \frac{1}{2} r \cdot r\big)\big]$.

Conditioning on $(u, \mathcal{B}, \mathcal{C}, \varepsilon)$:

$$p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) \propto \underbrace{p(\mathcal{B}, \mathcal{C} \mid w, u, \varepsilon)}_{\text{(ii)}} \cdot \underbrace{p(w \mid u)}_{\text{(i)}}$$

(i) C.3: $w \in \mathcal{C}$ implies the slice constraint holds $\Rightarrow p(w \mid u) \propto 1$ on $\mathcal{C}$.

(ii) C.2: current state $\in \mathcal{C}$; **C.4**: probability of generating $(\mathcal{B}, \mathcal{C})$ is the same for all $w \in \mathcal{C} \Rightarrow p(\mathcal{B}, \mathcal{C} \mid w, u, \varepsilon) \propto \mathbb{I}[w \in \mathcal{C}]$.

Substituting: $p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) \propto \mathbb{I}[w \in \mathcal{C}]$, i.e. uniform over $\mathcal{C}$.

By **C.1** (leapfrog is volume-preserving), all elements of $\mathcal{C}$ carry equal mass:

$$p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) = \frac{1}{|\mathcal{C}|}\, \mathbb{I}[w \in \mathcal{C}]. \qquad \square$$

Heuristic for Initial Epsilon

Algorithm 4 Heuristic for choosing an initial value of ϵ

function FindReasonableEpsilon(θ)

Initialize $\epsilon = 1, r \sim \mathcal{N}(0, I)$. ($I$ denotes the identity matrix.)

Set $\theta', r' \leftarrow$ Leapfrog(θ, r, ϵ).

$a \leftarrow 2\mathbb{I}\left[\frac{p(\theta',r')}{p(\theta,r)} > 0.5\right] - 1$.

while $\left(\frac{p(\theta',r')}{p(\theta,r)}\right)^a > 2^{-a}$ **do**

  $\epsilon \leftarrow 2^a\epsilon$.

  Set $\theta', r' \leftarrow$ Leapfrog(θ, r, ϵ).

end while

return ϵ.

Algorithm 5 Hamiltonian Monte Carlo with Dual Averaging

Given θ^0, δ, λ, $\mathcal{L}$, M, M^{adapt}:

Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta), \mu = \log(10\epsilon_0), \bar{\epsilon}_0 = 1, \bar{H}_0 = 0, \gamma = 0.05, t_0 = 10, \kappa = 0.75$.

for $m = 1$ to M **do**

  Reample $r^0 \sim \mathcal{N}(0, I)$.

  Set $\theta^m \leftarrow \theta^{m-1}, \tilde{\theta} \leftarrow \theta^{m-1}, \tilde{r} \leftarrow r^0, L_m = \max\{1, \text{Round}(\lambda/\epsilon_{m-1})\}$.

  for $i = 1$ to L_m **do**

    Set $\tilde{\theta}, \tilde{r} \leftarrow \text{Leapfrog}(\tilde{\theta}, \tilde{r}, \epsilon_{m-1})$.

  end for

  With probability $\alpha = \min\left\{1, \frac{\exp\{\mathcal{L}(\tilde{\theta}) - \frac{1}{2}\tilde{r}\cdot\tilde{r}\}}{\exp\{\mathcal{L}(\theta^{m-1}) - \frac{1}{2}r^0\cdot r^0\}}\right\}$, set $\theta^m \leftarrow \tilde{\theta}, r^m \leftarrow -\tilde{r}$.

  if $m \leq M^{\text{adapt}}$ **then**

    Set $\bar{H}_m = \left(1 - \frac{1}{m+t_0}\right)\bar{H}_{m-1} + \frac{1}{m+t_0}(\delta - \alpha)$.

    Set $\log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma}\bar{H}_m, \log \bar{\epsilon}_m = m^{-\kappa}\log \epsilon_m + (1 - m^{-\kappa})\log \bar{\epsilon}_{m-1}$.

  else

    Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.

  end if

end for

NUTS with Dual Averaging (1/2)

Algorithm 6 No-U-Turn Sampler with Dual Averaging (Main Loop)

Given θ^0, δ, $\mathcal{L}$, M, M^{adapt}:
Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1}) - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\theta^m = \theta^{m-1}$, $n = 1$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, _, _, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
    else
      $_, _, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
    end if
    if $s' = 1$ **then**
      With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
    end if
    $n \leftarrow n + n'$.
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  if $m \leq M^{\text{adapt}}$ **then**
    Set $\bar{H}_m = \left(1 - \frac{1}{m+t_0}\right) \bar{H}_{m-1} + \frac{1}{m+t_0}(\delta - \frac{\alpha}{n_\alpha})$.
    Set $\log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \epsilon_m + (1 - m^{-\kappa}) \log \bar{\epsilon}_{m-1}$.
  else
    Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
  end if

NUTS with Dual Averaging (2/2) - BuildTree

Algorithm 6 (continued)

function BuildTree($\theta, r, u, v, j, \epsilon, \theta^0, r^0$)
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow$ Leapfrog($\theta, r, v\epsilon$).
  $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  $s' \leftarrow \mathbb{I}[u < \exp\{\Delta_{\max} + \mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2} r^0 \cdot r^0\}\}, 1$.
else
  Recursion—implicitly build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow$ BuildTree($\theta, r, u, v, j-1, \epsilon, \theta^0, r^0$).
  if $s' = 1$ **then**
    if $v = -1$ **then**
      $\theta^-, r^-, _, _, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow$ BuildTree($\theta^-, r^-, u, v, j-1, \epsilon, \theta^0, r^0$).
    else
      $_, _, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow$ BuildTree($\theta^+, r^+, u, v, j-1, \epsilon, \theta^0, r^0$).
    end if
    With probability $\frac{n''}{n'+n''}$, set $\theta' \leftarrow \theta''$.
    Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
    $s' \leftarrow s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$
    $n' \leftarrow n' + n''$
  end if
  return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end if

Appendix: Benchmarks Bayesian & Hierarchical Logistic Regression Efficiency

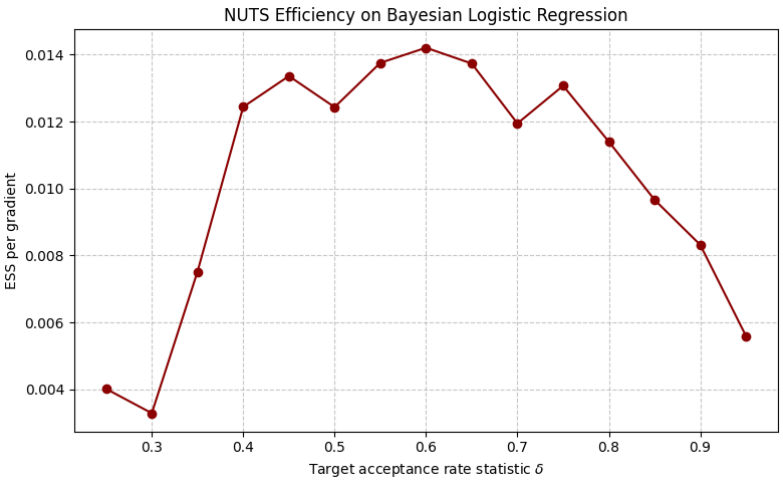


Figure 2: Bayesian Logistic Regression (25-D).

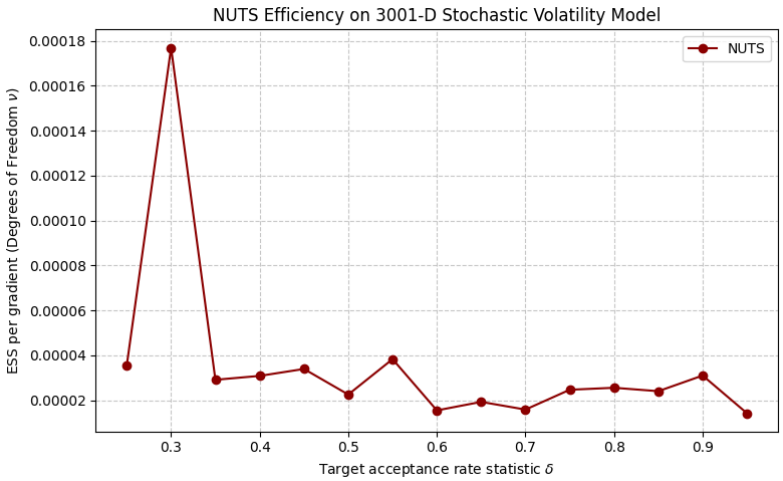


Figure 3: Hierarchical Logistic Regression (302-D).

Appendix: Dual Averaging Convergence

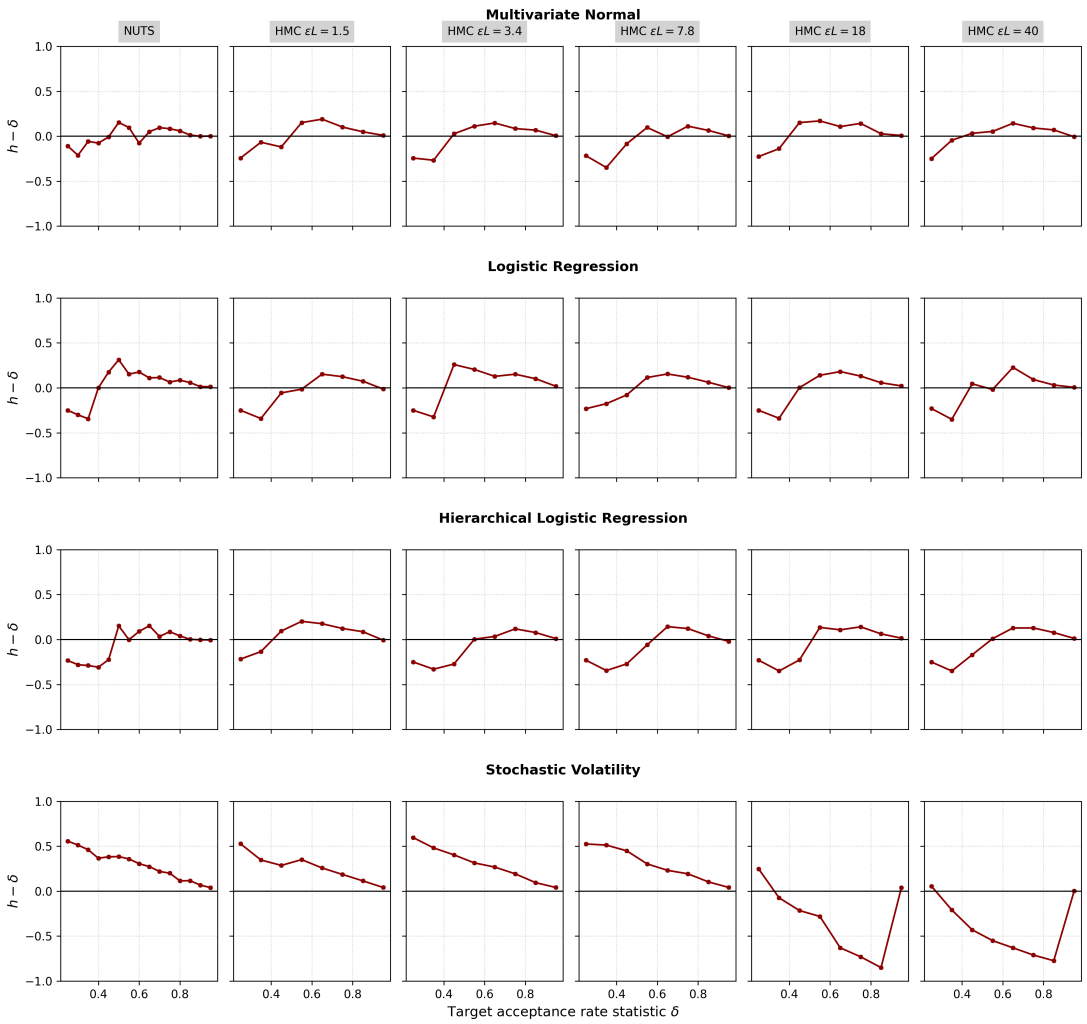


Figure 4: Discrepancy plots ($h - \delta$ across δ) for all four original benchmarks. Dual averaging reliably drives acceptance to target δ within a few hundred warm-up iterations.

Evaluating NUTS: The Paper's Benchmarks

The paper tests NUTS on four targets spanning low to very high dimensions, measuring **minimum ESS per gradient evaluation** across all dimensions.

250-D	25-D	302-D	3001-D
Multivariate Normal	Logistic Regression	Hierarchical LR	Stochastic Volatility

NUTS matches or exceeds best-tuned HMC on **all four benchmarks** without any trajectory-length tuning, up to 2× on the Gaussian and 1.5× on the stochastic volatility model.

But all four targets are **unimodal or mildly structured**. How does NUTS fare on posteriors with genuinely complex geometry?

Beyond the Paper: Three Stress Tests

We probe three failure modes of HMC that are absent from the original evaluation:

Multimodality

BNN, 191-D

$5! \times 5! =$

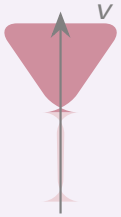
14400 modes

Dense Correlations

GP Classif., 150-D

dense 150×150 cov.

Varying Geometry



Neal's Funnel, 10-D

spatially varying σ

Stress Test 1: Bayesian Neural Network (191-D)

Setup: BNN $(30 \to 5 \to 5 \to 1)$, $\mathcal{N}(0, 100)$ priors on all weights, UCI Breast Cancer dataset. Weight-permutation symmetries create $5! \times 5! = 14{,}400$ equivalent modes.

NUTS explores a **wider region of weight space**; HMC stagnates in a single mode.

NUTS fails to achieve higher ESS per gradient than a well-tuned HMC (0.03×10^{-4} vs. 1.57×10^{-4} for HMC in our tests).

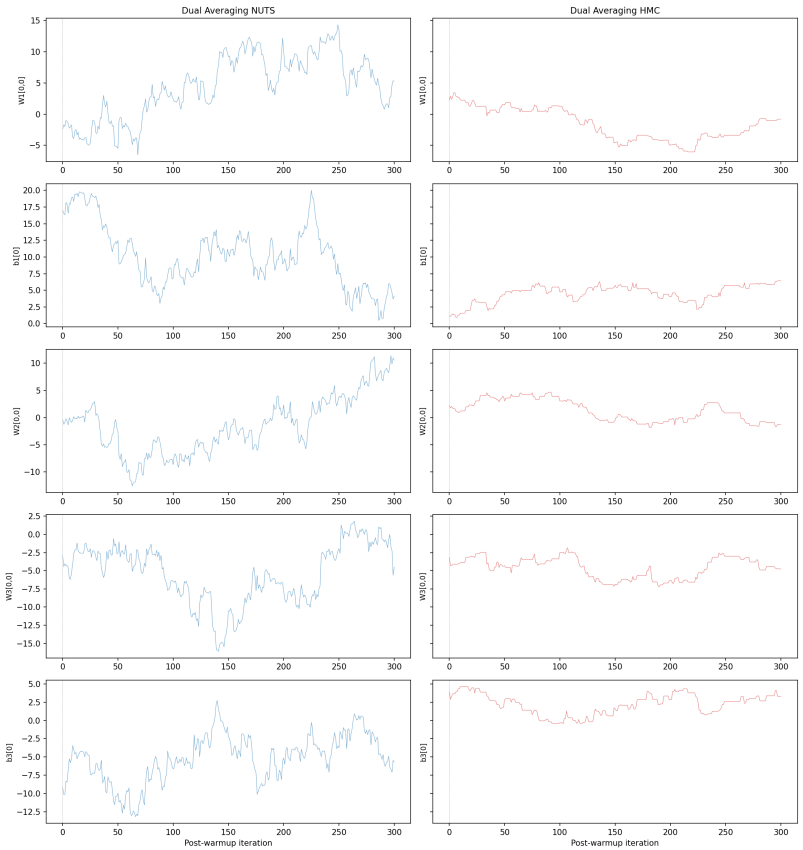


Figure 5: Trace plots: NUTS (blue) covers a broader range of weight values than HMC (red).

Stress Test 2: GP Classification (150-D)

Setup: RBF kernel ($\ell = 1$, $\sigma_f^2 = 1$), 150 California Housing points, fully dense K. The gradient $\nabla \log p(f) = -K^{-1} f + y \odot \sigma(-y \odot f)$ requires a dense solve at every leapfrog step.

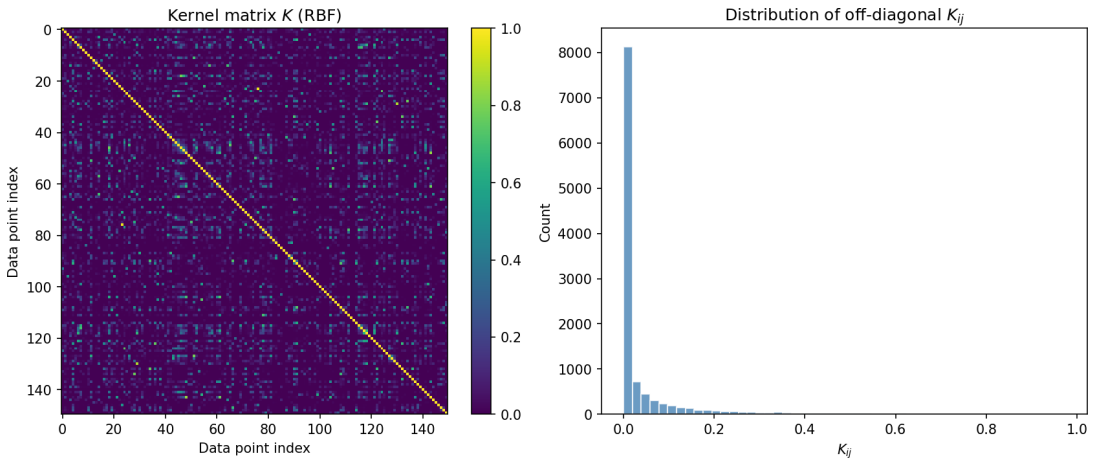


Figure 6: RBF kernel matrix K: dense correlation structure.

Nuanced result

NUTS: 2.14×10^{-3} ESS/grad HMC: 1.52×10^{-3} ESS/grad

Modest 40% gain, but NUTS uses $4\times$ more gradient evaluations. The dense covariance induces long trajectories before a U-turn is detected. The advantage over well-tuned HMC is narrow here.

Stress Test 3: Neal's Funnel (10-D)

Setup: A pathological geometry with a narrow neck and wide top:

$$v \sim \mathcal{N}(0, 9), \qquad x_i \mid v \sim \mathcal{N}(0, e^v), \quad i = 1, \ldots, 9$$

The challenge

$v \to -\infty$: $e^v \to 0$, needs tiny ε

$v > 0$: wide top, needs large ε

HMC's **fixed step size** cannot handle both.

Result

NUTS recovers $\mathrm{Var}(v) \approx 9$ (truth).

HMC is confined and severely underestimates $\mathrm{Var}(v)$.

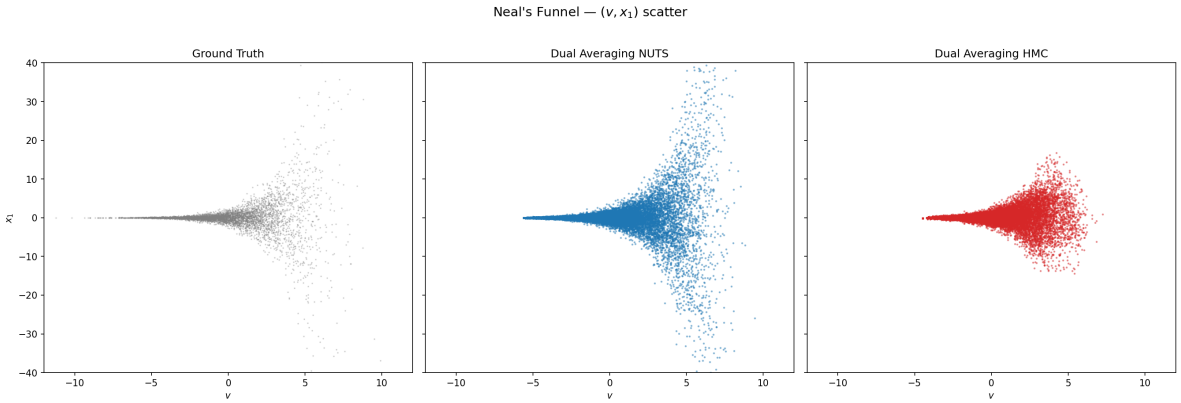


Figure 7: Left: ground truth. **Centre:** NUTS covers the full funnel. **Right:** HMC fails to fully explore the funnel.

Funnel: Trace Plots & Marginal

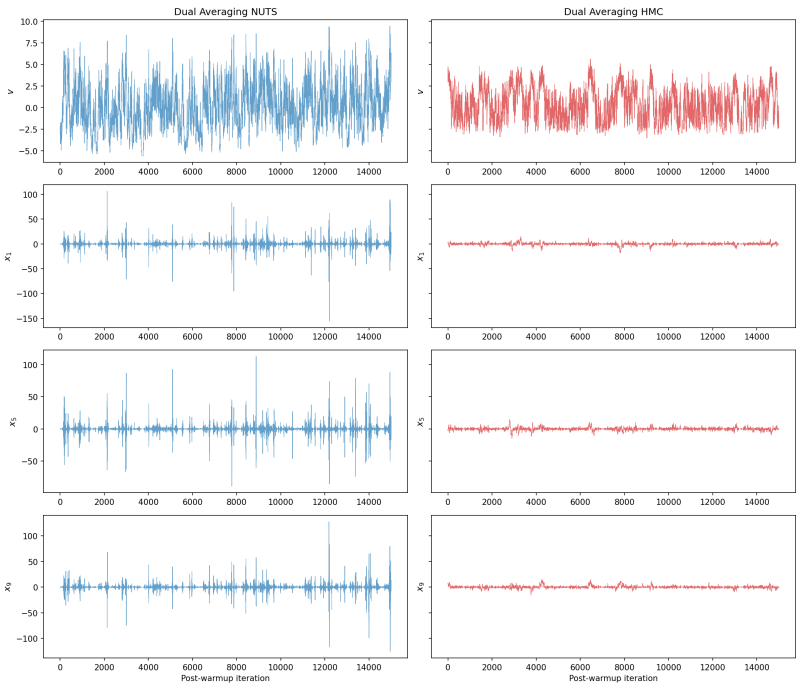


Figure 8: NUTS mixes freely; HMC shows flat segments (rejected proposals).

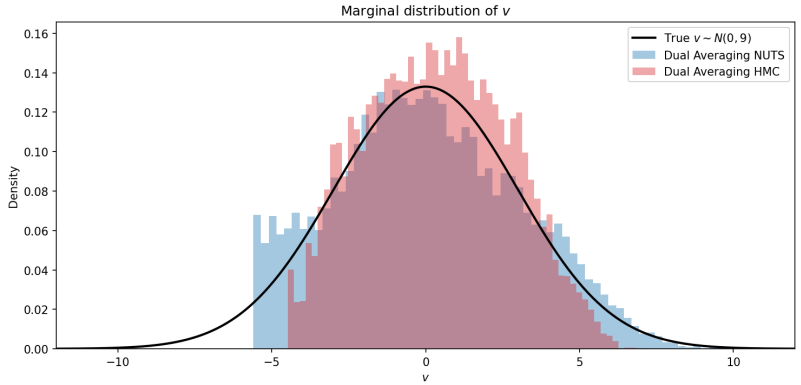


Figure 9: Marginal of v: NUTS recovers the truth; HMC misses $v < 0$ entirely.

Appendix: BNN Trace Plots

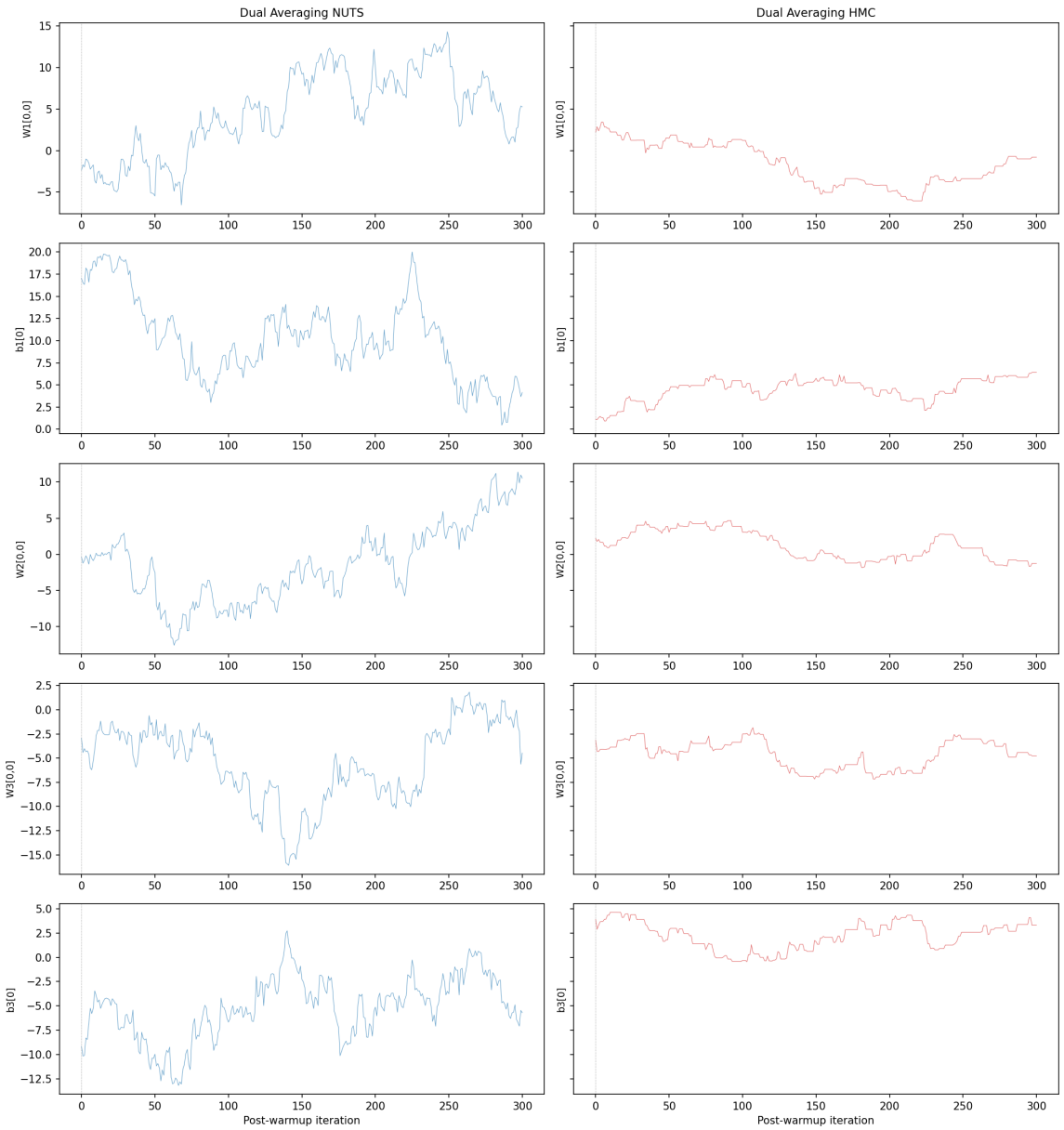


Figure 10: Trace plots: NUTS (blue) covers a broader range of weight values than HMC (red).

Appendix: BNN Adaptation Diagnostics

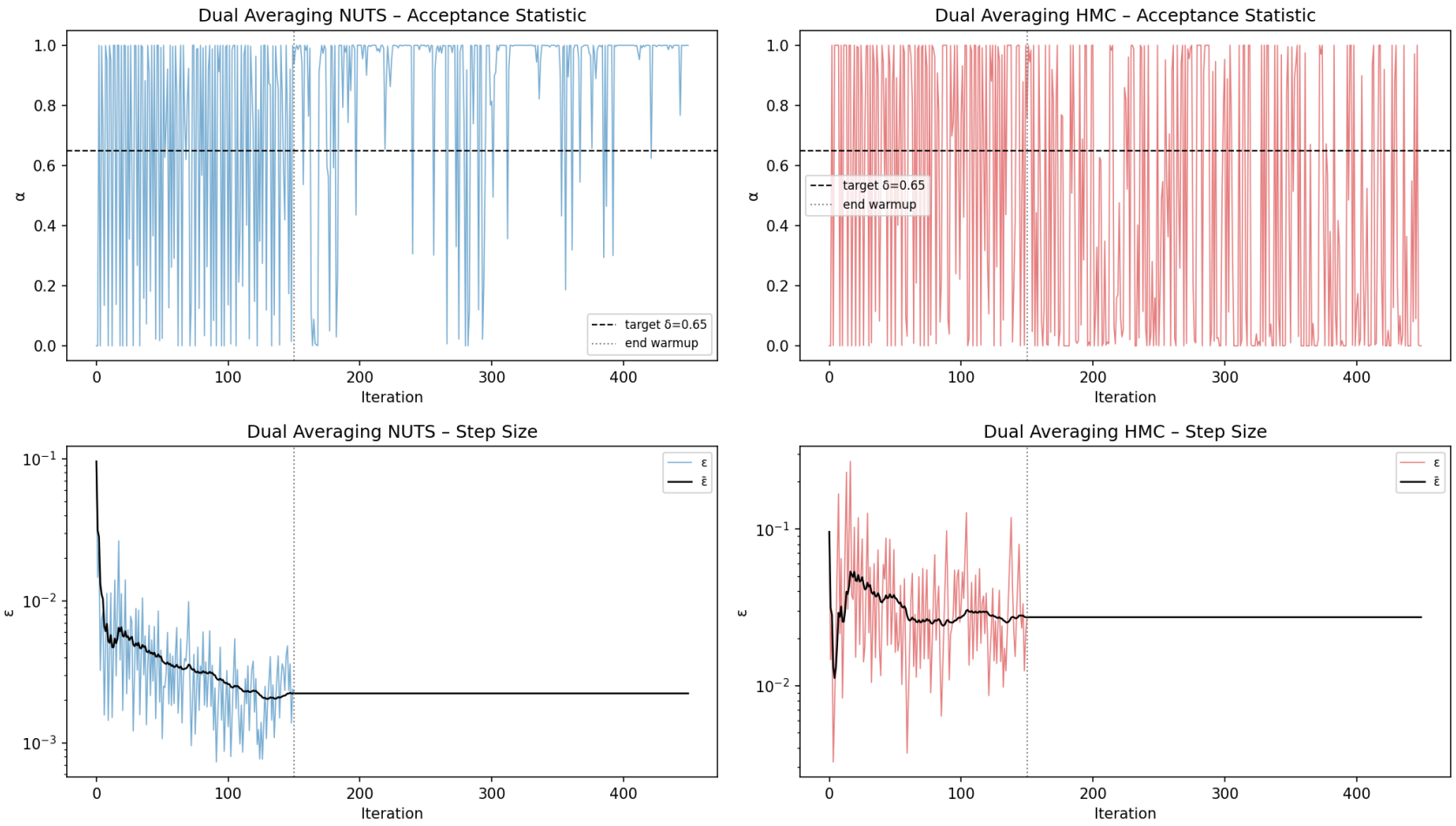


Figure 11: Dual-averaging convergence on the BNN. Step size $\bar{\varepsilon}$ stabilises within the warm-up phase.

BNN: Efficiency

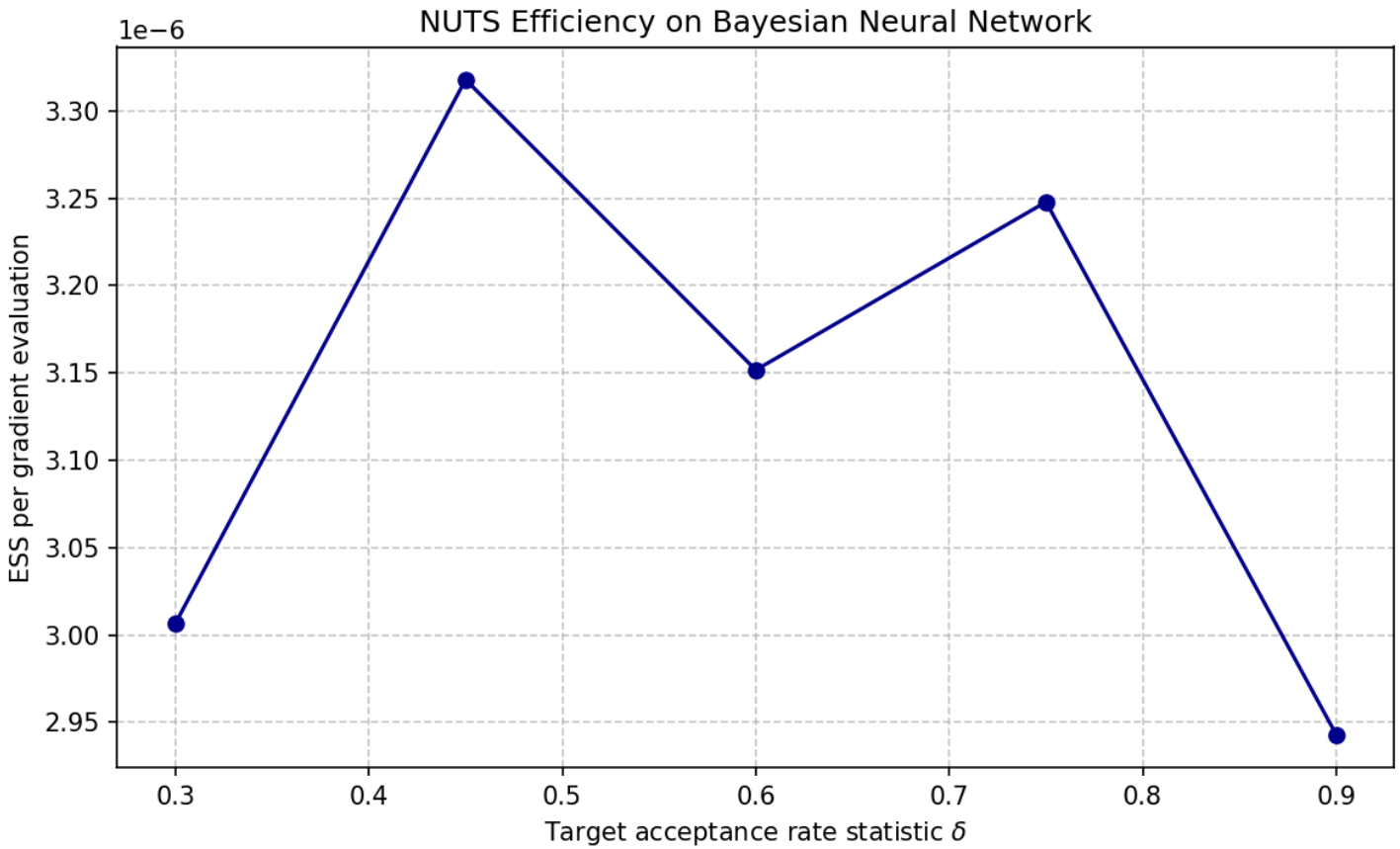


Figure 12: ESS/gradient on the BNN. NUTS consistently achieves higher effective samples per gradient evaluation than HMC across all tested configurations.

Appendix: GPC Kernel Matrix & Posterior Correlations

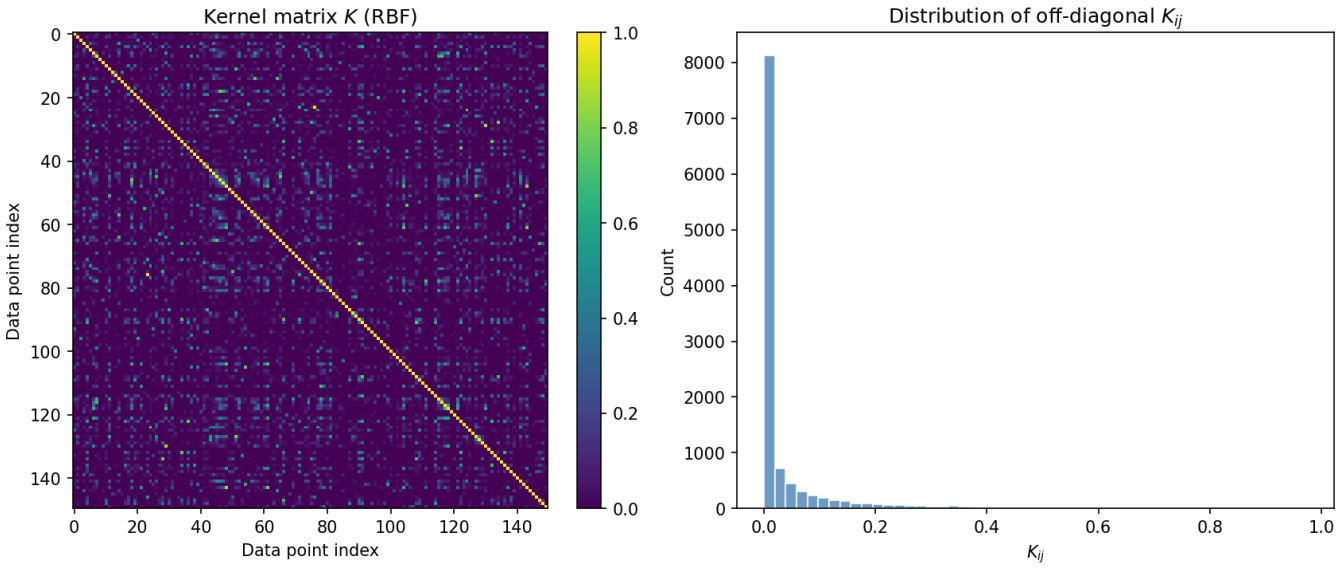


Figure 13: RBF kernel matrix K: dense correlation structure.

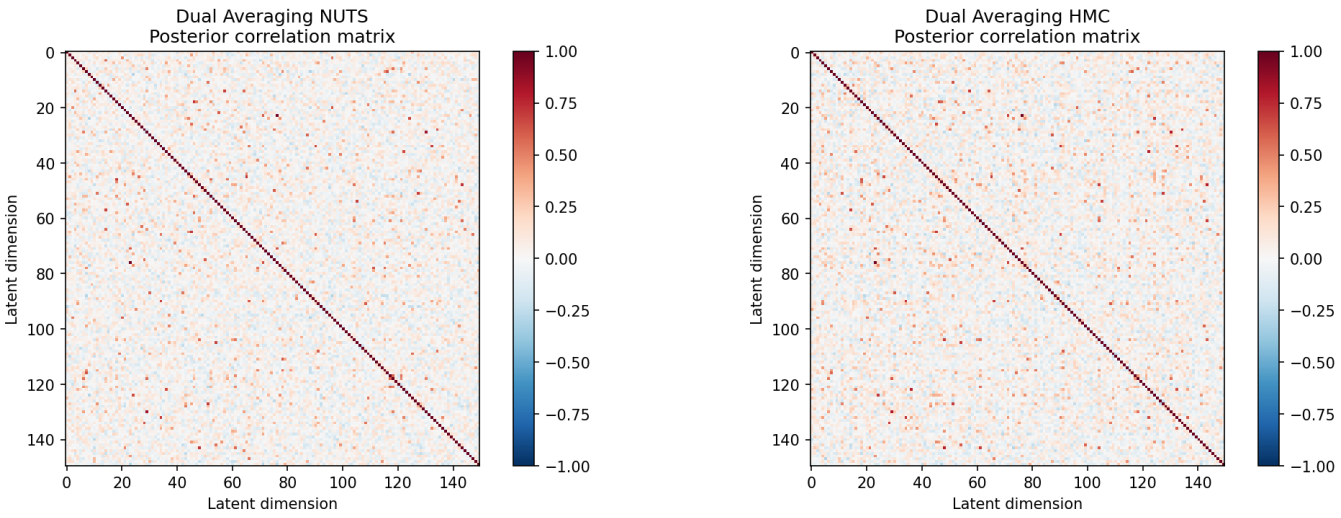


Figure 14: Posterior correlations among latent values f.

Appendix: GPC Efficiency

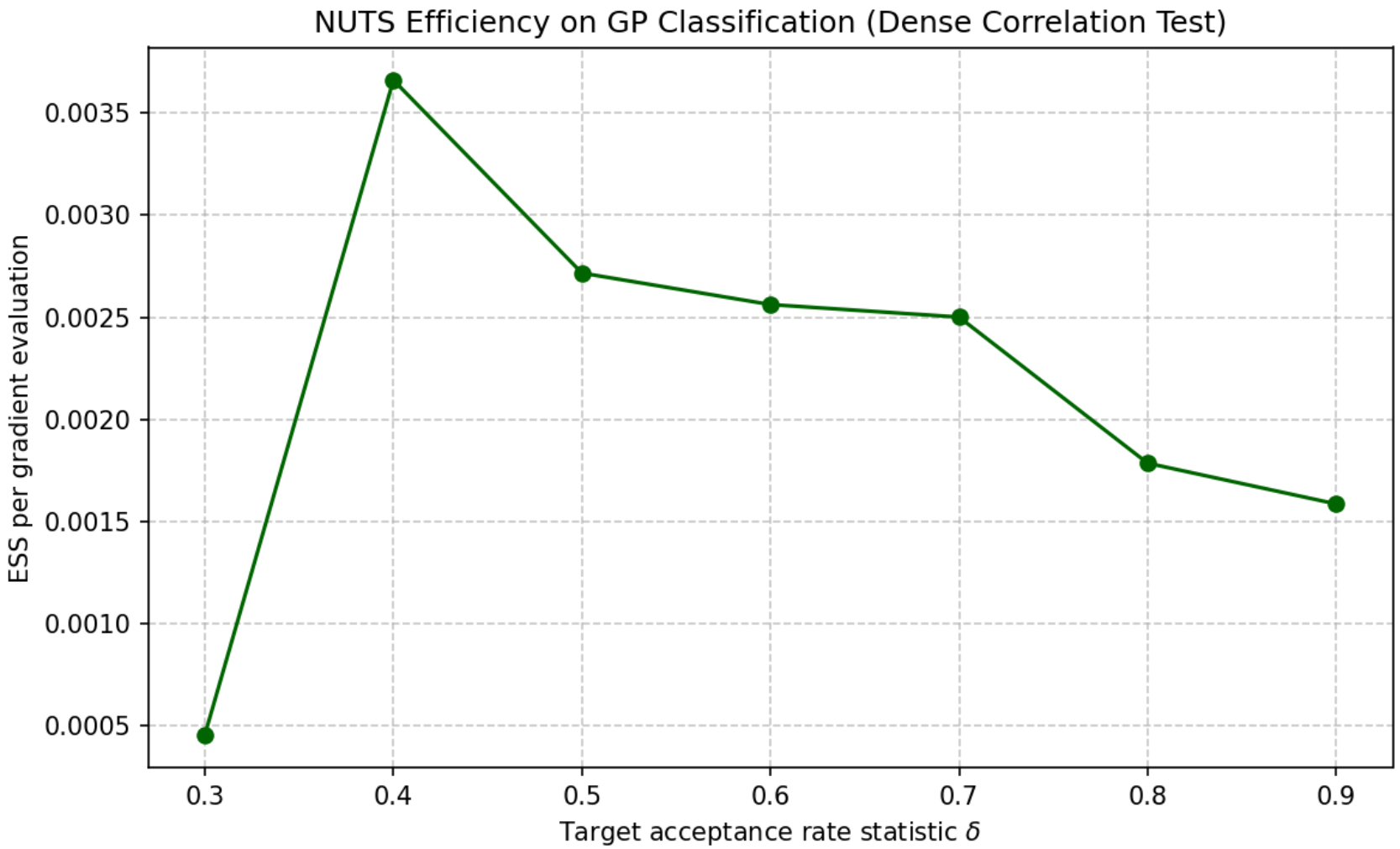


Figure 15: ESS/grad on GP classification. NUTS wins on per-gradient quality but at higher absolute cost.

Appendix: Funnel Scatter

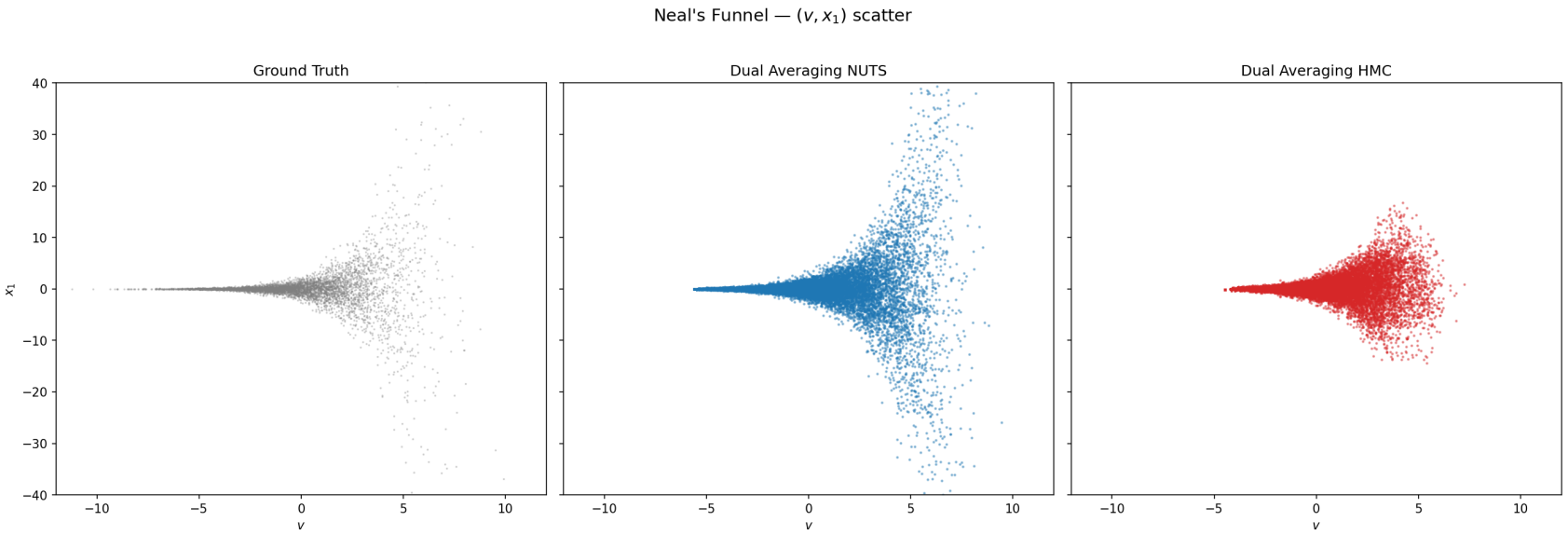


Figure 16: Left: ground truth. **Centre:** NUTS covers the full funnel. **Right:** HMC fails to explore it.

Appendix: Funnel Trace Plots & Marginal

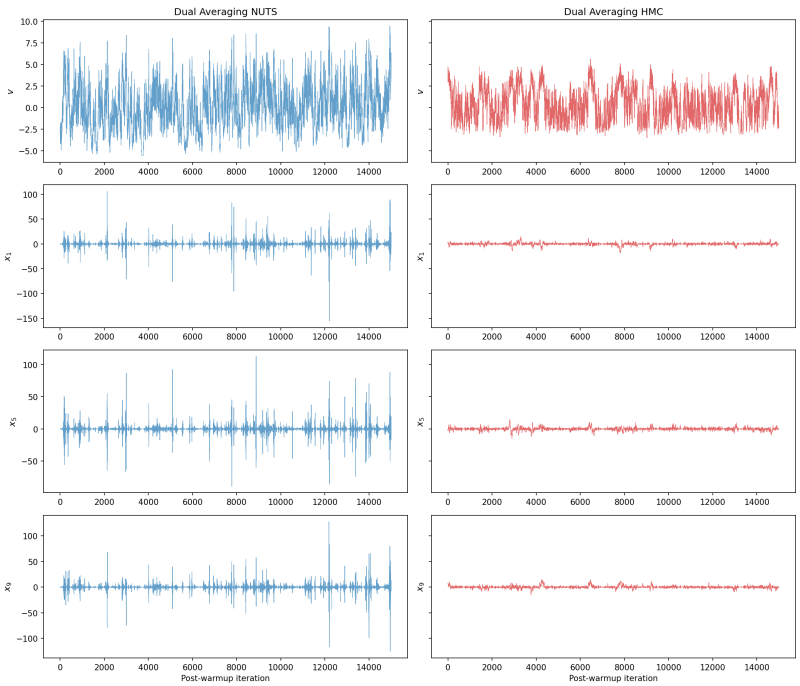


Figure 17: NUTS mixes freely; HMC shows flat segments.

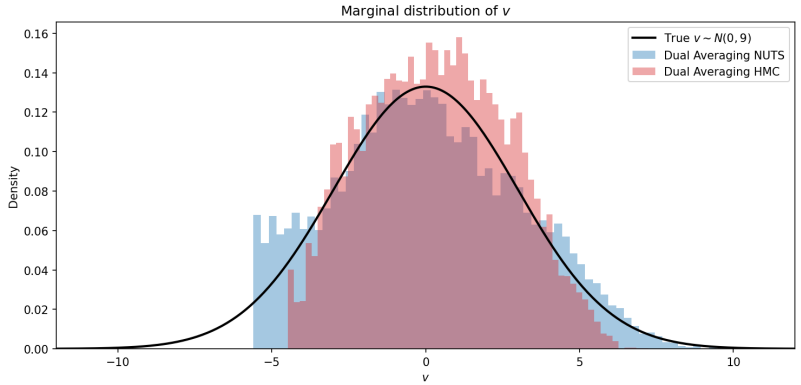


Figure 18: Marginal of v: NUTS recovers truth; HMC misses $v < 0$.

Appendix: Funnel Trajectory Lengths

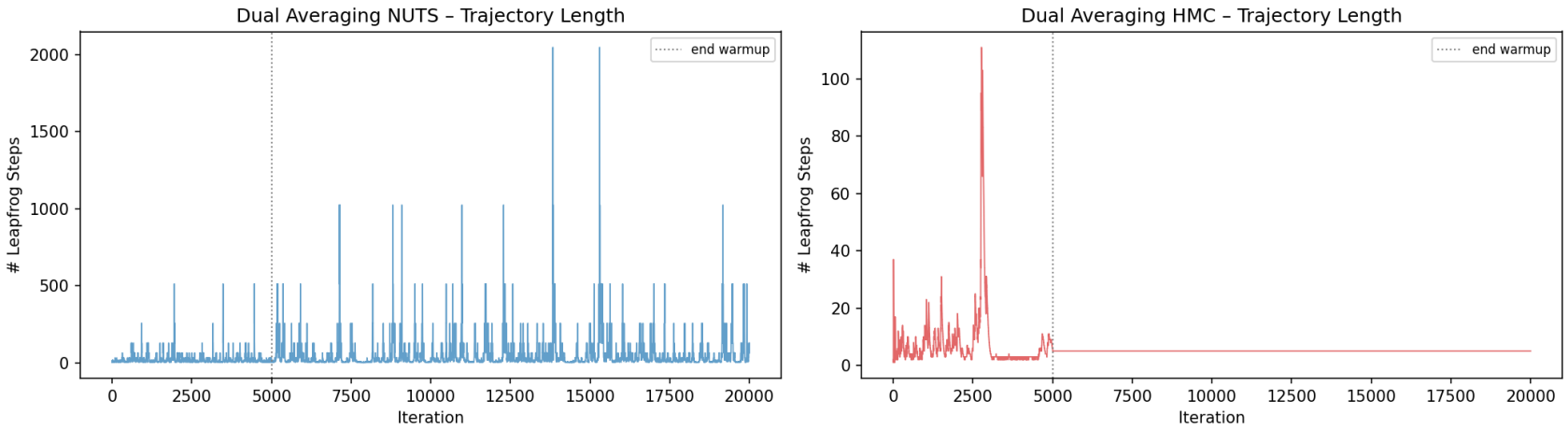


Figure 19: NUTS trajectory lengths on Neal's Funnel. Adaptive mechanism adjusts to local geometry.